

- 2 (a) (i) accelerations (A to B and B to C) are same magnitude B1
 accelerations (A to B and B to C) are opposite directions B1
 or both accelerations are toward B B1
 (A to B and B to C) the component of the weight down the slope provides the acceleration B1 [3]
- (ii) acceleration = $g \sin 15^\circ$ C1
 $s = 0 + \frac{1}{2} at^2$ C1
 $s = 0.26 / \sin 15^\circ = 1.0$
- $t^2 = \frac{1.0 \times 2}{9.8 \times \sin 15^\circ}$ A1
 $t = 0.89 \text{ s}$ [3]
- (iii) $v = 0 + g \sin 15^\circ t$ or $v^2 = 0 + 2g \sin 15^\circ \times 1.0$ C1
 $v = 2.26 \text{ m s}^{-1}$ A1 [2]
 (using loss of GPE = gain KE can score full marks)
- (b) loss of GPE at A = gain in GPE at C or loss of KE at B = gain in GPE at C B1
 $h_1 = h_2 = 0.26 \text{ m}$ or $\frac{1}{2} mv^2 = mgh$ $h_2 = 0.5 \times (2.26)^2 / 9.81 = 0.26 \text{ m}$
 $x = 0.26 / \sin 30^\circ = 0.52 \text{ m}$ A1 [2]

- 3 (a) power is the rate of doing work or power = work done / time (taken) or